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Insights into the thermochemistry of paracetamol in ternary mixtures of eutectic solvents

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Abstract
Solution enthalpies of paracetamol, $\Delta_{sol}H$, in choline chloride–ethylene glycol–water mixtures were evaluated at 298.15 K. Solvatochromic UV–vis shifts of five different probes (4-nitroaniline, 4-nitroanisole, 4-nitrophenol, *N*,*N*-dimethyl-4-nitroaniline and 2,6-diphenyl-4-(2,4,6-triphenylpyridinium-1-yl)phenolate) were also assessed for the same mixtures at the same temperature in order to compute the corresponding Kamlet–Taft parameters, α, β, and π^*. Quantitative structure–property relationships (QSPR) between $\Delta_{sol}H$ and the solvatochromic parameters were established to investigate the solution processes under consideration. Due to the small variability in the solvent parameters, a further set of eight pure solvents were considered. The behavior of paracetamol in eutectic mixtures is shown to be not significantly different from its behaviour in the pure solvents. The solvents' basicity measured by β, is the most relevant factor modeling the solution process of this solute in the total set of solvents.

Keywords Ternary mixture · Choline chloride · Ethylene glycol · Water · Kamlet–Taft parameters · Solution enthalpies

Introduction

The search for new solvents to replace common organic solvents with high toxicity and volatility, which lead to the dispersion of volatile organic compounds into the atmosphere, remains a current topic. In the last two decades, ionic liquids (ILs) have attracted great interest in the scientific community. Ionic liquids are molten salts, that are liquid at room temperature, and their properties can be adjusted by combining different cations and anions. Such solvents can be used, for example, in biocatalytic processes and extraction and electrochemical applications [1]. Over time, many researchers have considered ILs as a new class of green solvents since they can present low biodegradability and are considered biocompatible. However, more recently, there has been increasing concern about this supposed "greenness" given the results from more toxicity studies [2, 3].

A promising alternative to ILs are eutectic solvents (ESs), whose properties can, in many cases, be fine-tuned to become more environmentally friendly and are, in general, much easier and cheaper to obtain. ESs have already demonstrated great potential for use in various applications such as syntheses, gas absorption, and biocatalytic processes [1, 4].

ESs are described as the combination of two or more pure components, solid or liquid, in which one of them is a hydrogen bond acceptor (HBA), usually a salt, and the other is a hydrogen bond donor (HBD), for example, an alcohol, an amine, or an acid. The mixture formed by these two components has a melting point that is lower than that of the pure components caused in large part by hydrogen bonds formed during the preparation of the mixture [4–6].

The melting temperature of the ideal eutectic mixture ($T_{E,ideal}$) does not result from a linear combination of the melting temperatures of the starting compounds, $T_{fus,A}$ and $T_{fus,B}$, at a given composition. As shown in Fig. 1, there is a greater deviation (negative deviation from ideality) of the eutectic temperature ($T_{E\ real}$, blue line) from the ideal eutectic temperature ($T_{E,ideal}$). The eutectic mixture remains

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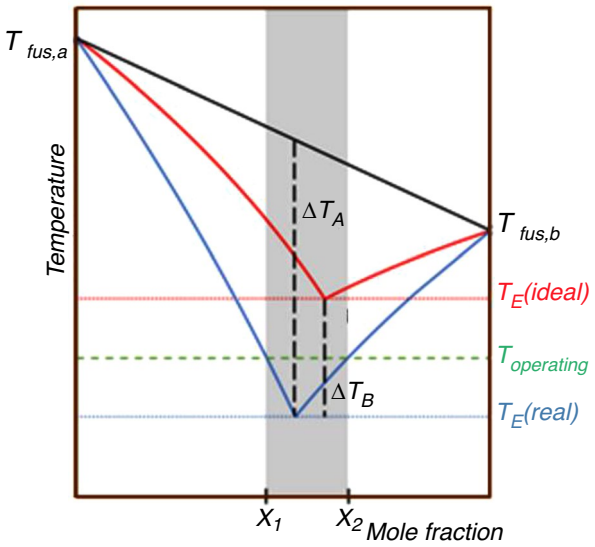


Fig. 1 Phase diagram of a binary eutectic mixture. The red line represents an ideal eutectic mixture and the blue line a deep eutectic mixture [6]

liquid, due to the large depression in the melting temperature, for a given range of compositions (x_1-x_2), typically at room temperature [6, 7]. To distinguish a "deep eutectic solvent" (DES) from other eutectic mixtures and to give meaning to the term "deep," it is necessary to define it as a mixture of pure compounds where the eutectic point temperature is lower than that of an ideal liquid mixture. This temperature difference should be defined as the discrepancy (ΔT_B) between the ideal eutectic point temperature (T_{Eideal}) and the actual eutectic point temperature ($T_{E,real}$), rather than the difference (ΔT_A) between the melting points of the individual components and the actual eutectic point temperature.

A disadvantage of many studied ESs compared to ILs is that the former are less chemically inert and, in many cases, show higher viscosity. One of the most common ways to deal with the viscosity issue is to use a third component—a co-solvent, such as water—without altering other properties [1, 5, 7].

One of the promising areas for which eutectic mixtures are being studied is their use as alternative solvents to conventional organic solvents for the synthesis of different active pharmaceutical ingredients (APIs). These eutectic solvents have also shown promise in drug delivery systems due to their ability to enhance drug solubility and stability [8]. Additionally, the use of DESs comprising active pharmaceutical ingredients (THEDES) has been proposed as a means to improve therapeutic efficiency and drug delivery, with the potential to integrate them into (bio)polymer-based systems [9].

However, thermodynamic solution data for these systems are very scarce. Nevertheless, such data are essential

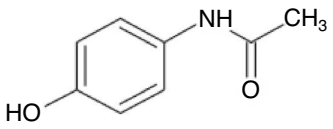


Fig. 2 Structural formula of paracetamol

for the proper industrial use of these solvents and for a better understanding of the nature of these mixtures at the molecular level.

In this study, we have focused our attention on paracetamol (Fig. 2, IUPAC name *N*-(4-hydroxyphenyl) acetamide).

Paracetamol is an over-the-counter analgesic and antipyretic medication used for mild to moderate pain and fever. It is harmless in low doses, but when taken in high doses, it can cause acute liver damage and death due to liver failure. Paracetamol is a white powder, slightly soluble in cold water, with a p*K*a value of 9.50 [10].

Solution enthalpy, $\Delta_{sol}H$, is a very sensitive property to solvent effects and has been used to evaluate and quantify the various molecular interactions involved in solution processes. In these studies, infinite dilution enthalpies of solution, $\Delta_{sol}H^{\infty}$, are used, as this allows solute–solute interactions to be disregarded, with attention focused exclusively on solute–solvent interactions [11].

The solution enthalpy process can be described based on three energetic contributions: (i) an endothermic contribution corresponding to the breaking of solute–solute interactions, (ii) an endothermic contribution relating to the creation of a cavity of adequate size to accommodate the solute, which will correspond to the breaking of solvent–solvent interactions, and (iii) an exothermic interaction, which accounts for the accommodation of the solute in the formed cavity and subsequent formation of specific and non-specific solute–solvent interactions and the restoration of solvent–solvent interactions.

In general, if the energy required to form solute–solvent interactions is greater than the sum of the energies required to destroy solute–solute and solvent–solvent interactions, the solution process will be exothermic. Otherwise, the solution process will be endothermic.

However, the significance of the solution enthalpy being either endothermic or exothermic extends beyond the theoretical understanding of bond-breaking and bond-forming processes since practical aspects like solubility and stability are dependent on the sign of the enthalpy variation. An exothermic dissolution typically indicates better solubility at lower temperatures, whereas an endothermic process suggests improved solubility at higher temperatures. This is important for predicting and controlling dissolution rates for various applications, including pharmaceutical

ones, where the rate of drug dissolution can affect the drug's bioavailability.

Although the aim of this work is to make an enthalpy study, it is important to retain that entropic factors are also involved in dissolution. Entropic contributions, which are related to the degree of disorder or randomness in the system, play a crucial role in determining the spontaneity of the dissolution process. Even in cases where the enthalpy change is unfavorable (endothermic), a significant increase in entropy upon dissolution can drive the process forward.

In this work, solution enthalpies were determined in a solution calorimeter operating under semi-adiabatic conditions, with the thermostatic bath temperature maintained constant at 25.00 °C throughout all the experiments.

To better understand the solvation process as a whole, it is necessary to understand the interaction mechanisms between solute and solvent molecules at a molecular level. One way to do this is through the use of solvatochromic probes, a valuable tool for measuring the polarity of liquid media, not only because they allow the identification of different types of interactions, but also because they allow the quantification of those interactions [11]. The polarity of a solvent can be defined as the solvation power of the solvent, and it depends on the intervention of all specific and non-specific intermolecular interactions between solute ions or molecules and solvent molecules, except for those interactions leading to chemical modifications of solutes [12, 13].

Solvatochromism is related to the change in position, intensity, and shape of the absorption bands of appropriate probes due to the influence of probe–solvent interactions that mimic the same type of interactions occurring between solute and solvent. These interactions influence the degree of stabilization of the ground state and the first excited states of the probe molecule, causing an increase or decrease in the energy differential. This leads to a blue or red shift, respectively, with increasing of solvent polarity [14].

Given the different types of interactions usually present in solution processes, several probes are needed to mirror each type of specific or non-specific interaction. Based on these probes, different scales have been built to measure the magnitude of those interactions, as will be seen below.

Among the variety of scales proposed over the years, the most successful ones are undoubtedly the Dimroth and Reichardt's $E_T(30)$, reflecting a blend of dipolarity and Lewis acidity, and the scales proposed by Kamlet and Taft for non-specific (dipolarity/polarizability) interactions, π^*, and also Lewis basicity, β, and Lewis acidity, α. The $E_T(30)$ scale relies on the molar transition energy of the probe 2,6-diphenyl-4-(2,4,6-triphenylpyridinium-1-yl)phenolate, commonly known as betaine (30), in a liquid medium, calculated on the basis of the wavelength value, λ, of its absorption maximum—Eq. (1) [11, 12, 15].

$$E_T(30) = \frac{28590\ \text{nm}}{\lambda_{\text{betaine (30)}}} \tag{1}$$

Kamlet and Taft scales used in this work are the ones later on perfected by Laurence and Nicolet, who advocated the advantages of using a reduced number of probes, namely 4-nitroanisole, 4-nitroaniline, 4-nitrophenol, *N*,*N*-dimethyl-4-nitroaniline, and, again, betaine (30) dye (or betaine (33))—Fig. 3 [12, 14–16].

Betaine (30) is a molecule whose solvatochromic bands are in the visible region and therefore the variation in the band position with solvent polarity is easily noticeable. However, the E_T scale has some limitations, the main ones being the low solubility of betaine (30) in some nonpolar solvents for which a more lipophilic penta-t-butyl-substituted dye is used, and the use of more acidic solutions for which a less basic dichloro-substituted dye, 2,6-dichloro-4-(2,4,6-triphenyl-1-pyridinium-1-yl)phenolate or betaine (33), is the best choice—Fig. 3f [17]. Since tabulated ET values for various solvents are available using both betaine (30) and betaine (33) probes, and a good correlation exists between them, it is possible to determine ET(30) values from ET(33) values and vice versa. [17].

The parameter $E_T(30)$ was subsequently normalized to make it dimensionless, leading to the parameter $E_T^N(30)$ calculated from Eq. (2), where the less polar solvent tetramethylsilane (TMS, $E_T^N = 0.00$), and the most polar solvent water ($E_T^N = 1.00$) were used as reference solvents[17]:

$$E_T^N = \frac{E_T(30) - 30.7}{32.4} \tag{2}$$

The π^* parameter is a measure of all non-specific intermolecular interactions, providing a mixed measure of solvent dipolarity and polarizability [12]. Laurence and Nicolet proposed the use of two probes to measure these interactions, 4-nitroanisole and *N*,*N*-dimethyl-4-nitroaniline, which led to two sub-scales of π^*, π^*_{OMe}, and π^*_{NMe2}, respectively. The values of these sub-scales are obtained through Eqs. 3 and 4, where the wavenumber, $\bar{\nu}$, is expressed in kilo Kaiser ($1\text{kK} = 10^3\ \text{cm}^{-1}$). These two sub-scales were normalized using cyclohexane, where $\pi^* = 0$, and dimethyl sulfoxide (DMSO), where $\pi^* = 1$, as reference solvents [11, 12].

$$\pi^*_{\text{OMe}} = \frac{[\bar{\nu}(4-\text{nitroanisole})_{\text{Solvent}}] - 34.12}{(31.72 - 34.12)} \tag{3}$$

$$\pi^*_{\text{NMe2}} = \frac{[\bar{\nu}(N,N-\text{dimethyl}-4-\text{nitroaniline})_{\text{Solvent}}] - 28.18}{(24.66 - 28.18)} \tag{4}$$

The β parameter assesses the specific solute–solvent interactions related to the basicity of hydrogen bond acceptor solvents. Two probe pairs are used to calculate the β scale:

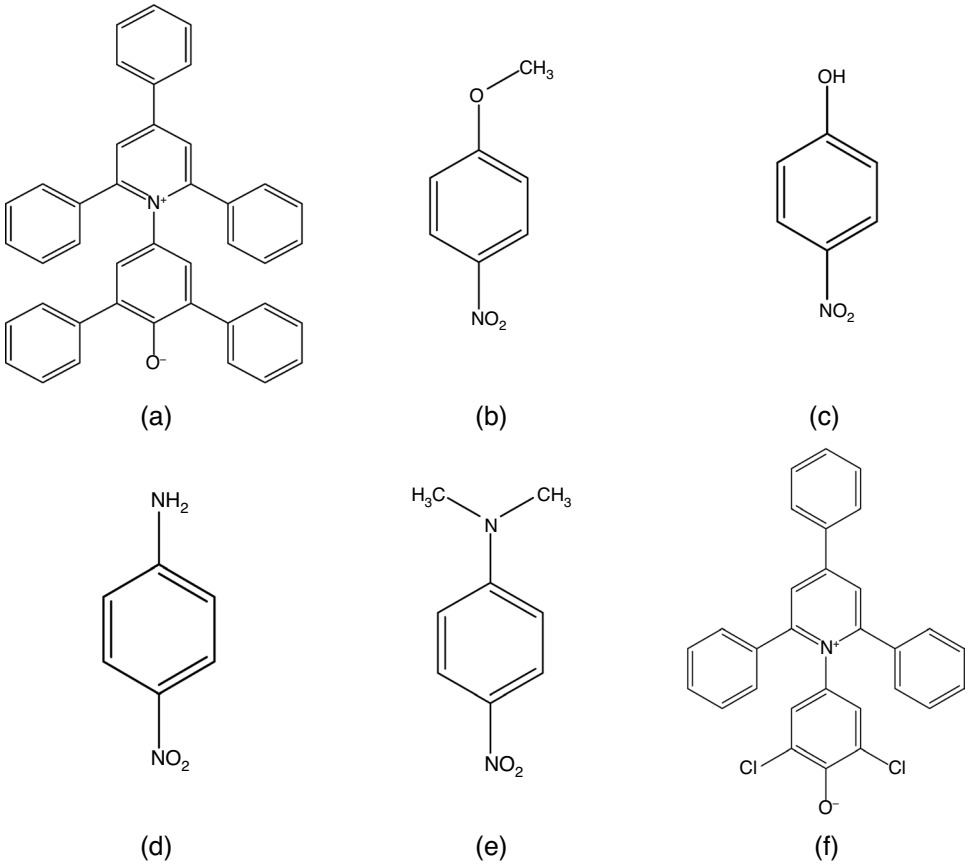


Fig. 3 **a** Betaine (30), **b** 4-nitroanisole, **c** 4-nitrophenol, **d** 4-nitroaniline, **e** *N*,*N*-dimethyl-4-nitroaniline, **f** Betaine (33)

4-nitrophenol and 4-nitrosole on one hand, and *N*,*N*-dimethyl-4-nitroaniline and 4-nitroaniline, on the other hand. The β_{OH} sub-scale is obtained from the first pair, where the hydrogen bond donor group is the -OH group, while the β_{NH2} sub-scale is obtained from the second pair, where the hydrogen bond donor group is the -NH group. Each sub-scale is calculated according to Eqs. 5 and 6. These scales, which are normally averaged, were normalized using hexamethylphosphoramide and cyclohexane as reference solvents, with β values of 1 and 0, respectively [16, 18].

The α scale quantifies specific solute–solvent interactions related to the acidity of hydrogen bond donor solvents, compared to solute–methanol interactions for which $\alpha = 1$. Two sub-scales were developed using the betaine (30), which is sensitive to solvent acidity, and either 4-nitrosole or *N*,*N*-dimethyl-4-nitroaniline, resulting in the α_{OMe} orα_{NMe2} sub-scales, respectively. Both sub-scales are computed using Eqs. 7 and 8, with the average of the two sub-scales (α_{avg}) being commonly used [11, 19, 20].

$$\beta_{OH} = \frac{[(1.0434 \times \overline{\nu}(4-\text{nitroanisole})_{Solvent}) - 0.57] - \overline{\nu}(4-\text{nitrophenol})_{Solvent}}{2} \quad (5)$$

$$\beta_{NH2} = \frac{[(0.9841 \times \overline{\nu}(N,N-\text{dimethyl}-4-\text{nitroaniline})_{Solvent}) + 3.49] - \overline{\nu}(4-\text{nitroaniline})_{Solvent}}{2.759} \quad (6)$$

$$\alpha_{OMe} = \frac{[(1.873 \times \overline{\nu}(4-\text{nitroanisole})_{Solvent}) - 74.58] + \overline{\nu}(\text{betaine (30)})_{Solvent}}{6.24} \quad (7)$$

$$\alpha_{\mathrm{NMe2}} = \frac{[(1.318 \times \overline{\nu}(\mathrm{N,N-dimethyl-4-nitroaniline})_{\mathrm{Solvent}} - 47.7] + \overline{\nu}(\mathrm{betaine\ (30)})_{\mathrm{Solvent}}}{5.47} \quad (8)$$

In general, these parameters are used collectively to establish multiple linear models, on one hand, with a predictive character toward systems that have not yet been experimentally assessed, and, on the other hand, to allow a physicochemical interpretation of the studied phenomenon. The models are called quantitative structure–property relationships (QSPR) since they establish a relationship between a chosen physicochemical property of a given system, and the set of molecular descriptors that encode the system's structural characteristics. QSPR includes linear and nonlinear relationships. The methodology for establishing these relationships may rely on multiple linear regression, partial least squares regression, artificial neural networks, and support vector machines, among others.

In this work, quantitative multilinear relationships between solution enthalpies and Kamlet–Taft parameters were obtained, to expose the dominant factors governing the studied solution processes.

Materials and methods

Reagents

The ternary and binary eutectic mixtures were prepared using choline chloride(Acros Organics, 99%), ethylene glycol (Sigma-Aldrich, 99.8%), both used without further purification, and MilliQwater (Millipore, resistivity 18.2 MΩ cm, at 25 °C). The chosen solvatochromic probes were as follows: betaine (30) (Aldrich, 90%), betaine (33) (Carbosynth, 97%), 4-nitroaniline (Aldrich, 99%), *N*,*N*-dimethyl-4-nitroaniline (TCI, 98%), 4-nitroanisole (TCI, 98%) and 4-nitrophenol (Merck, 99,5%). Paracetamol (Aldrich > = 98%) was used as API in the solution enthalpy studies in both eutectic mixtures and in the pure solvents depicted in Table 1.

Table 1 Pure organic solvents tested

Solvents	Source	Purity/%
Acetone	Honeywell	≥ 99.5
Acetonitrile	Honeywell	≥ 99.9
Nitromethane	Sigma-Aldrich	≥ 96
Dimethyl sulfoxide	Honeywell	≥ 99.9
N,*N*-Dimethylacetamide	Sigma-Aldrich	99
Methanol	Sigma-Aldrich	99.8
2-Propanol	AnalaR	≥ 95
2-Butanol	Merck	≥ 99

Mixture preparation

Mixtures were prepared by weighing each component in an Erlenmeyer, using a Precisa analytical balance (uncertainty of ± 0.0001 g). The obtained mole fractions are indicated in Table S1.

In choline chloride (ChCl) mixtures, the solid was first weighed and then placed in a drying oven for at least 24 h being weighed again after that time. Based on the final mass of ChCl, the required amount of the other components was then determined. Subsequently, the components were mixed in a low humidity environment (a nitrogen-saturated glove box) and heated to 373 K for one hour in a silicone bath with a slow and constant magnetic stirring until a homogeneous liquid was formed. Before use, the mixtures were allowed to reach ambient temperature. All binary and ternary mixtures containing choline chloride were tested for stability allowing them to rest for 24 h and then checking to verify if no solid was formed. Only mixtures which remained as a homogeneous liquid mixture after this period of time were subsequently tested.

Figure 4, shows the data points for the choline chloride + ethylene glycol + water (ChCl-Eg-H_2O) mixture prepared in this work, which remained stable after 24 hours. This figure also defines the stability zone for this mixture. The

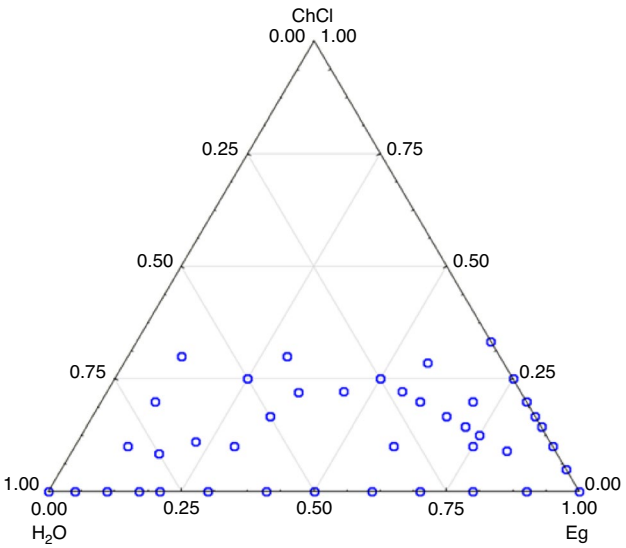


Fig. 4 ChCl-Eg-H_2O tested mole fraction mixtures

corresponding individual mole fractions tested are specified in Table S1 of supplementary material.

Solvatochromic parameters

UV–Vis spectroscopic analysis was performed at 298.15 K in a Thermo Fisher, Nicolet Evolution 300 spectrophotometer, equipped with a jacketed cell module which allowed a thermostatic control from a recirculating water bath with an uncertainty of ± 0.1 K. A pair of matched quartz analytical cells of 1 cm path length was used in all measurements.

Each solvatochromic probe was pre-dissolved in the mixtures, and the concentration was adjusted by adding more solvent mixture or solvatochromic probe to obtain absorbance values in the range 0.6–0.9.

The wavelengths of maximum absorbance, for the five dyes, resulted from at least an average of three individual experiments and always presented a standard deviation smaller than 0.1. The values for each individual mole fraction are presented in Tables S2–S7.

Solution enthalpies

Calorimetric measurements were carried out at 298.15 K using a Thermometric precision solution calorimeter, previously described [21]. The equipment includes a 100-mL Pyrex glass vessel fitted with a thermistor for sensing temperature and a heater for calibration and equilibration. The calorimeter was tested using the solution process of tris(hydroxymethyl)aminomethane(TRIS) in NaOH 0.05 mol.dm^{-3} and HCl 0.1 mol dm^{-3}. Experimental and literature values agreed within experimental uncertainty [22]. The solute (paracetamol) was also weighed in a Precisa analytical balance with a precision of ± 0.1 mg and its concentration ranged from 0.01 to 0.02 mol dm^{-3}. Cylindrical ampoules were filled with solute and sealed. Heats of empty ampoule breaking were found to be negligible. The reported enthalpy value for paracetamol in each mixture or pure solvent resulted from the average of at least three independent experiments, with a relative standard deviation always less than 5%. Paracetamol was only tested in selected mole factions of the ternary mixture ChCl-Eg-H_2O for reasons discussed ahead.

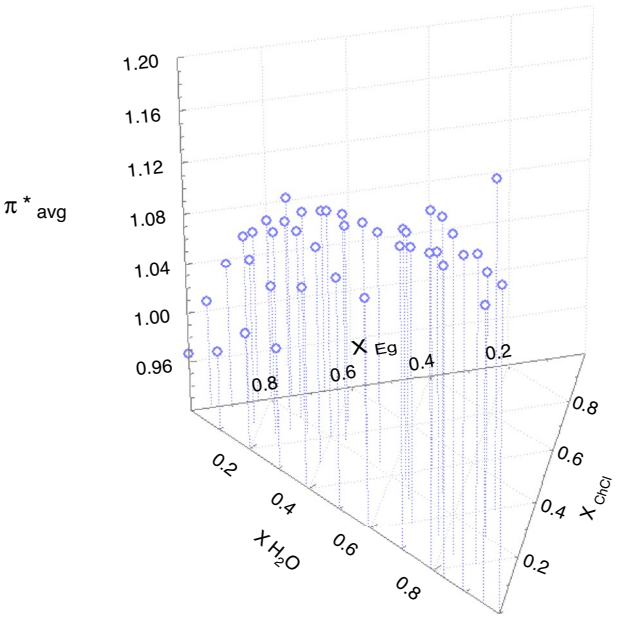


Fig. 5 Variation of π^*_{avg} in the ternary mixture ChCl-Eg-H_2O

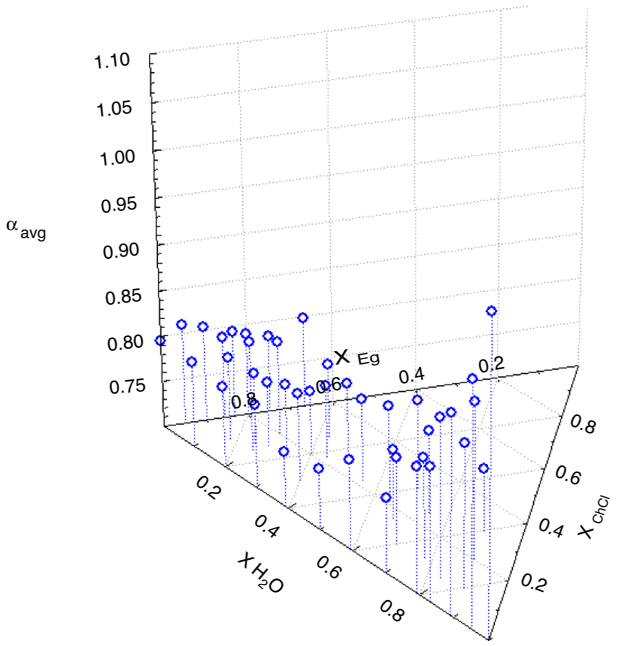


Fig. 6 Variation of α_{avg} in the ternary mixture ChCl-Eg-H_2O

Results and discussion

Solvatochromic and Kamlet–Taft parameters

Using the wavelength of the absorption maxima obtained for each of the solvatochromic probes at 298.15 K (Tables S2–S7), wavenumbers ($\overline{\nu} = 1/\lambda$) for each dye were computed. These values were then used to calculate the Kamlet–Taft parameters π^*, α and β, as described in the introduction.

The averaged values of the Kamlet–Taft parameters, π^*_{avg}, α_{avg} and β_{avg}, are summarized in Table S8 (see supplementary material). These results are also depicted in the ternary plots of ChCl-Eg-H_2O of Figs. 5, 6, 7.

Looking strictly at the binary fractions of the mixture ChCl-Eg, we can see that the studied properties do not vary

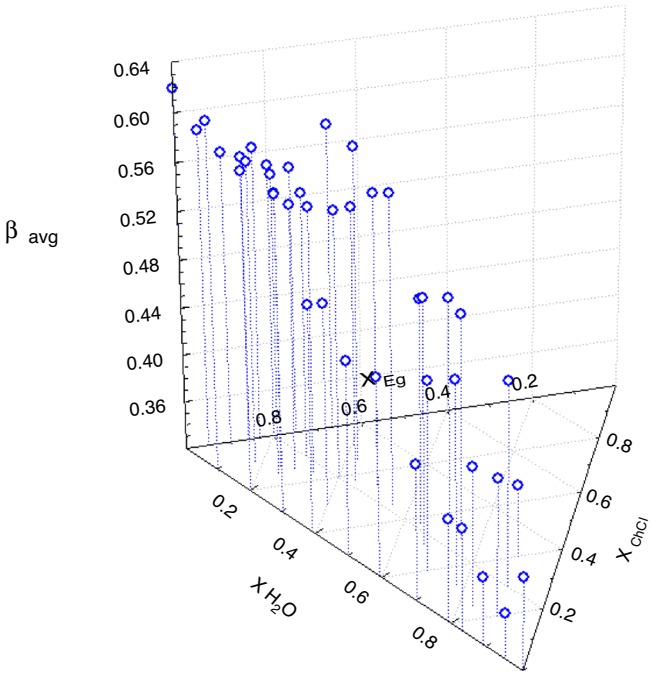


Fig. 7 Variation of β_{avg} in the ternary mixture ChCl-Eg-H_2O

significantly. The use of the non-eutectic mixture Eg-H_2O allowed a noteworthy increase in the attainable number of points and thus an increase in the variation range of the solvatochromic parameters. On the other hand, there are no data for the ChCl-H_2O mixture due to the insolubility of the probes in this solvent mixture.

The π^* parameter is the one with the least variation, and the small observed increment is positively correlated with the larger mole fractions of water, thus suggesting an increase in the capacity of mixtures with a higher percentage of water to disperse charges in the region around the probe.

The β parameter shows a maximum for pure ethylene glycol and it decreases with an increasing mole fraction of water. The variation of the α parameter is once again directly related to the water content of the mixture, increasing with the acidity of the solvent. The solvatochromic characterization of the ternary mixtures suggests, in turn, that the third component (water) plays, important role in solute–solvent interactions. This outcome is anticipated, as the binary mixture of Eg-H_2O exhibits strong non-additive trends, particularly in the α and π^* parameters, indicating significant intermolecular interactions between the components. These interactions manifest in properties such as negative excess molar volumes [23]. Consequently, similar behavior is expected in the ternary mixtures.

Table 2 Solution enthalpies of paracetamol in the ChCl-Eg-H_2O tested mole fractions

x_{ChCl}	x_{Eg}	x_{H2O}	$\Delta_{sol}H \pm s$ ($\Delta_{sol}H$)/kJ mol^{-1}
0.00	1.00	0.00	11.8 ± 0.4
0.00	0.00	1.00	19.3 ± 0.4
0.33	0.67	0.00	11.2 ± 0.5
0.25	0.75	0.00	11.4 ± 0.2
0.14	0.86	0.00	11.7 ± 0.3
0.00	0.05	0.95	21.5 ± 0.6
0.00	0.11	0.89	22.4 ± 0.9
0.00	0.17	0.83	22.6 ± 0.5
0.00	0.21	0.79	21.9 ± 0.6
0.00	0.40	0.60	18.3 ± 0.4
0.00	0.80	0.20	13.2 ± 0.6
0.25	0.50	0.25	13.6 ± 0.6
0.17	0.33	0.50	15.7 ± 0.7
0.10	0.10	0.80	18.8 ± 0.6
0.30	0.10	0.60	16.9 ± 0.8
0.30	0.30	0.40	14.5 ± 0.5

Solution enthalpies in the ternary mixture ChCl-Eg-H_2O

Based on the stability zone previously established for the mixtures and on prior solubility tests to certify that the paracetamol dissolution process was sufficiently fast, some

Table 3 Experimental solution enthalpies of paracetamol in pure organic solvents and Kamlet–Taft parameters obtained from various bibliographic references[a]

Solvent	$\Delta_{sol}H \pm s$ ($\Delta_{sol}H$)/kJ mol^{-1}	π^*	α	β
Acetone	13.4 ± 0.3	0.71	0.08	0.48
Acetonitrile	22.3 ± 0.2	0.75	0.19	0.37
Nitromethane	30.5 ± 0.8	0.85	0.22	0.25
Dimethylsulfoxide	0.45 ± 0.04	1.00	0.00	0.76
N,*N*-Dimethylacetamide	− 2.3 ± 0.2	0.85[b]	0.00	0.76
Methanol	13.0 ± 0.2	0.60	0.98	0.73
2-Propanol	11.2 ± 0.4	0.48	0.68	0.93
2-Butanol	11.8 ± 0.5	0.54	0.54	0.91

[a]From ref. [24] and references therein

[b]From ref [15]

Table 4 Determination coefficients (r^2) obtained for the various uniparametric correlations, excluding pure solvents ($N = 14$)

Parameters	r^2
α_{avg}	0.44
β_{avg}	0.82
π^*_{avg}	0.67

mole fractions were selected for calorimetric analysis. Table 2 shows the different mixtures studied, the respective enthalpy of solution values and the associated standard deviations.

Knowing that the stability limit in the binary mixtures of ChCl-Eg corresponds to a mole fraction of 0.33 for ChCl, we found that in these eutectic mixtures, increasing the mole fraction of ethylene glycol led to an increase in the enthalpy of solution. However, from a statistical point of view, the values are identical within the margin of uncertainty.

Also, the addition of water increases, in general, the values of the solution enthalpies of paracetamol, both in the ternary mixtures and the binary mixture Eg-Water.

Solution enthalpies in pure solvents

Due to the relatively small variation of some of the solvatochromic parameters and the need to use them in the subsequent data treatment, the solution enthalpies of paracetamol in eight additional pure organic solvents were also experimentally determined. This decision has also allowed a deeper understanding of the thermochemical behavior of paracetamol in the ternary mixture choline chloride–ethylene glycol–water and the evaluation of how this mixture compares to pure solvents.

Solution enthalpies of paracetamol in the pure organic solvents are depicted in Table 3, along with the solvatochromic Kamlet–Taft parameters obtained from other literature sources and previously used by our group [24].

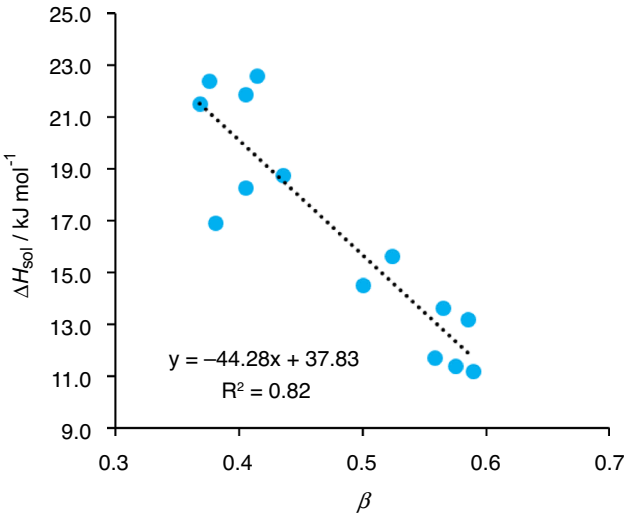


Fig. 8 Solution enthalpies of paracetamol in ChCl-Eg-H_2O mixtures as a function of β.

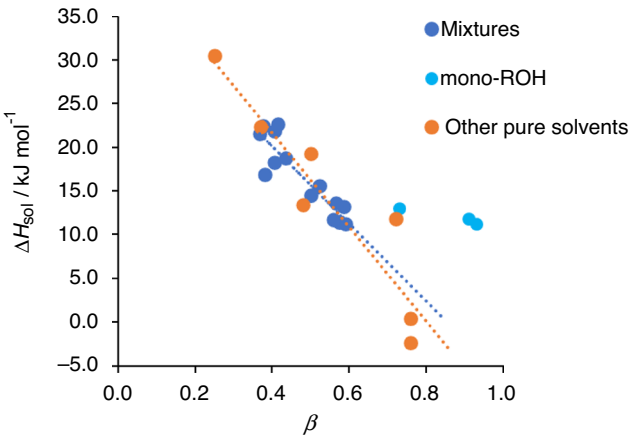


Fig. 9 Correlation between solution enthalpies and β for eutectic mixtures and pure solvents

Correlation analysis

Experimental results were analyzed using a QSPR approach. A QSPR model equation relates a given property of a system and a set of molecular descriptors that encode particular structural characteristics of that system—Eq. (8). This methodology has been previously used by our group to study several kinetic and equilibrium properties, among which, the solution enthalpies of different solutes in pure and mixed solvents [25–30].

$$\text{Property} = f(\text{molecular descriptors}) + \text{error} \quad (8)$$

As mentioned earlier, the molecular descriptors obtained in this work were the Kamlet–Taft solvatochromic parameters (α, π^*, β). Starting with the uniparametric correlations, it was immediately evident that the best one (*i.e.*, that with the highest coefficient of determination) was obtained with the β parameter—Table 4 and Fig. 8.

From Fig. 8, one can see that the higher the β, the lower the enthalpies of solution (*i.e.*, the less endothermic is the process). However, only 82% of the variability of the system is explained by this uniparametric model which reveals that this parameter alone is not sufficient to explain the solution process of paracetamol in these solvents.

In the process of checking for similarities between pure solvents and eutectic mixtures, a correlation was also established between the enthalpies of solution in the pure tested solvents and β. The large deviation observed for monoalcohols led to the elimination of these solvents from the regression and to a new refit with the remaining solvents.

Figure 9 shows that the lines corresponding to the two correlations are practically overlapping, which points out to a remarkably close behavior between the two sets of solvents, described in both cases predominantly by β. Moreover, the equation that results from considering simultaneously all solvents, *i.e.*, eutectic mixtures and pure solvents excluding the three monoalcohols, is as follows—Eq. (9):

$$\Delta_{sol}H = -\underset{(100\%)}{(49.6 \pm 4.6)}\beta + \underset{(100\%)}{(40.6 \pm 2.4)}$$
$$N = 21;\ r^2 = 0.86;\ s_{adj} = 2.8;\ F = 114 \quad (9)$$

As only 86% of the $\Delta_{sol}H$ variability is explained by β, multiparametric equations involving the remaining solvatochromic parameters were also tested. All biparametric regressions, *i.e.*, regressions with all double combinations of β, π^* and α, were tested and none was as successful as the one with β alone.

The triparametric regression was also verified, leading to a determination coefficient of 0.85—Eq. (10)—with monoalcohols showing once again a clear deviation from the trend line,

$$\Delta_{sol}H = -\underset{(100\%)}{(19.7 \pm 3.9)}\pi * + \underset{(100\%)}{(9.7 \pm 2.1)}\alpha$$
$$- \underset{(100\%)}{(43.1 \pm 4.4)}\beta + \underset{(100\%)}{(51.5 \pm 5.3)} \quad (10)$$
$$N = 24;\ r^2 = 0.85;\ s_{adj} = 2.9;\ F = 38$$

The exclusion of the three monoalcohols finally leads to the following equation and figures of merit—Eq. (11):

$$\Delta_{sol}H = -\underset{(100\%)}{(13.5 \pm 5.6)}\pi * + \underset{(100\%)}{(8.1 \pm 2.4)}\alpha$$
$$- \underset{(100\%)}{(48.2 \pm 3.9)}\beta + \underset{(100\%)}{(48.5 \pm 5.3)} \quad (11)$$
$$N = 21;\ r^2 = 0.91;\ s_{adj} = 2.3;\ F = 61$$

The analysis of Eq. (11) shows that, as expected, β is the dominant parameter, contributing negatively to $\Delta_{sol}H$, that is, making the solution process more exothermic. This exothermic contribution was already discussed in previous occasions by our group for other hydroxylic solutes (*e.g.*, 1-adamantanol) and is explained by the acidic characteristics of the solute's hydroxyl group [26, 27]. In turn, the dipolarity/polarizability of the solvent (π^*) also makes the process more exothermic (negative coefficient), while the solvent's hydrogen bond donor capacity, explained by α, makes it more endothermic.

Although a better correlation has been reached with the triparametric equation, the variability in $\Delta_{sol}H$ is still not fully explained by this set of parameters. The introduction of an appropriate solvent structuring term may eventually improve the model as already observed in previous studies [27, 30]. Typically, descriptors like the Hildebrand parameter which is related to the cohesive energy density (obtained from the standard molar enthalpy of vaporization and the molar volume) have been used in the past to describe solvent–solvent interactions. However, the estimation of this descriptor for solvent mixtures, and particularly for eutectic solvent mixtures, is still not well established and needs further investigation.

Conclusions

The Kamlet–Taft parameters obtained for the binary and ternary mixtures studied in this work did not show a great variability in the binary mixture ChCl-Eg. The solvatochromic parameters vary with the addition of water producing a decrease in β and an increase in α, while π^* presents the smallest variation, being the little increment positively correlated with larger amounts of water.

Solution enthalpies of paracetamol in the binary mixtures ChCl-Eg also showed little variability which increased again with the addition of water making them more endothermic. To expand the parameters' variability and assess their effect on the enthalpies of solution, eight additional pure solvents were added to the initial set. A QSPR analysis between $\Delta_{sol}H$ and the Kamlet–Taft parameters for all the solvents revealed that the best correlation

involved α, β and π^*, being β the most relevant term, with a negative contribution to $\Delta_{sol}H$, due to the acidic character of paracetamol. However, in spite of the good statistical quality of the regression, the introduction of a structuring descriptor would still probably improve the model's overall robustness. Also, this study showed that the behavior of these eutectic mixtures is not significantly different from that of pure solvents and can be modeled exactly by the same set of descriptors.

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